

Assignment 2

November 2025

1 REINFORCE on LunarLander-v2

Objective

The goal of this exercise is to implement the REINFORCE algorithm and analyze its performance on the `LunarLander-v2` environment.

T1. Implementation of REINFORCE

Open the notebook `04 LAB A Reinforce.ipynb` and implement the REINFORCE algorithm by completing the provided sections. In particular, you are required to define the policy network, specify the hyperparameters, and implement the training loop.

T2. Theoretical Questions and Coding Implementation

Answer the following questions in Markdown within the notebook and complete the required coding tasks:

1. Why is REINFORCE considered a Monte Carlo method?
2. What is the main drawback that REINFORCE suffers from, and what is the role of a control variate (baseline) in this context?
3. Implement in your code at least one variance-reduction strategy and compare the learning curves with and without this modification. Briefly comment on the observed differences.

T3. Learning Curve Analysis and Hyperparameter Summary

In your notebook, include the following:

- A plot showing the episode reward obtained throughout the training process.

Additionally, provide a table summarizing the main hyperparameters used in your final implementation. The table should include:

- learning rate,
- number of training episodes,
- discount factor γ ,
- maximum number of steps per episode,
- batch size (if applicable),
- network architecture (e.g., number of layers and units per layer).

You are also encouraged to report alternative hyperparameter configurations you tested (e.g., different learning rates or network sizes) and briefly discuss how these choices affected training stability and performance.

2 Actor–Critic on CartPole-v1

Objective

The goal of this exercise is to implement an Actor–Critic method, combining a policy network (actor) with a value function (critic).

T1. Implementation of Actor–Critic

Open the notebook `05 LAB_A Actor Critic.ipynb` and implement an Actor–Critic algorithm. Your implementation should include:

- the policy network (actor),
- the value network (critic),
- the temporal-difference (TD) error,
- the actor update based on the advantage signal.

T2. Theoretical Questions and Code Explanation

In a Markdown section of the notebook, address the following points:

- Explain the key differences between REINFORCE and Actor–Critic methods.
- Describe the roles of the actor, critic, and advantage in the learning process, indicating where the TD error is computed and how it is used in both updates.

Whenever possible, relate your explanation to specific code blocks in the notebook, briefly discussing how the theoretical concepts are reflected in your implementation.

T3. Learning Curve Analysis and Hyperparameter Summary

In your notebook, include the following:

- A plot showing the episode reward obtained throughout the training process.

Additionally, provide a table summarizing the main hyperparameters used in your final implementation. The table should include:

- learning rate,
- number of training episodes,
- discount factor γ ,
- maximum number of steps per episode,
- batch size (if applicable),
- network architecture (e.g., number of layers and units per layer).

You are also encouraged to report alternative hyperparameter configurations you tested (e.g., different learning rates or network sizes) and briefly discuss how these choices affected training stability and performance.

3 Proximal Policy Optimization (PPO)

Objective

The goal of this exercise is to train a PPO agent on a standard control environment and analyze its learning behaviour under different hyperparameter configurations.

T1. PPO with Stable-Baselines3

Choose one environment from the following options:

- `CartPole-v1`,
- `LunarLander-v2`,
- any other environment of your choice from the `gymnasium`.

Using either your own implementation of PPO from scratch or the PPO class from `stable-baselines3`, train an agent on the selected environment. Clearly specify:

- the policy network architecture,
- the main hyperparameters (learning rate, number of steps per update, batch size, total timesteps, discount factor γ , etc.).

T2. Theoretical Questions on PPO

Answer the following theoretical questions in Markdown in your notebook. When appropriate, relate your explanation to concepts or operations that appear in the PPO implementation.

1. Is PPO an on-policy or off-policy method? Explain what this means and discuss the implications for how data must be collected and used during training.
2. Explain the meaning of the policy ratio used in PPO. What does it represent, and why is it important for the update rule?
3. Describe the purpose of the clipping operation in PPO and explain how it helps prevent excessively large policy updates.
4. The PPO objective may include an entropy bonus. Explain the role of entropy regularization and how it influences the agent's exploration behaviour.

T3. Learning Curves and Hyperparameter Analysis

In your notebook, include:

- a plot of the episode rewards (or a smoothed version) over the course of training;
- a table summarizing the hyperparameters used in your *final* configuration;
- a short discussion of at least two alternative configurations you tried (e.g., different learning rates, batch sizes, or network sizes) and how they affected stability and performance.